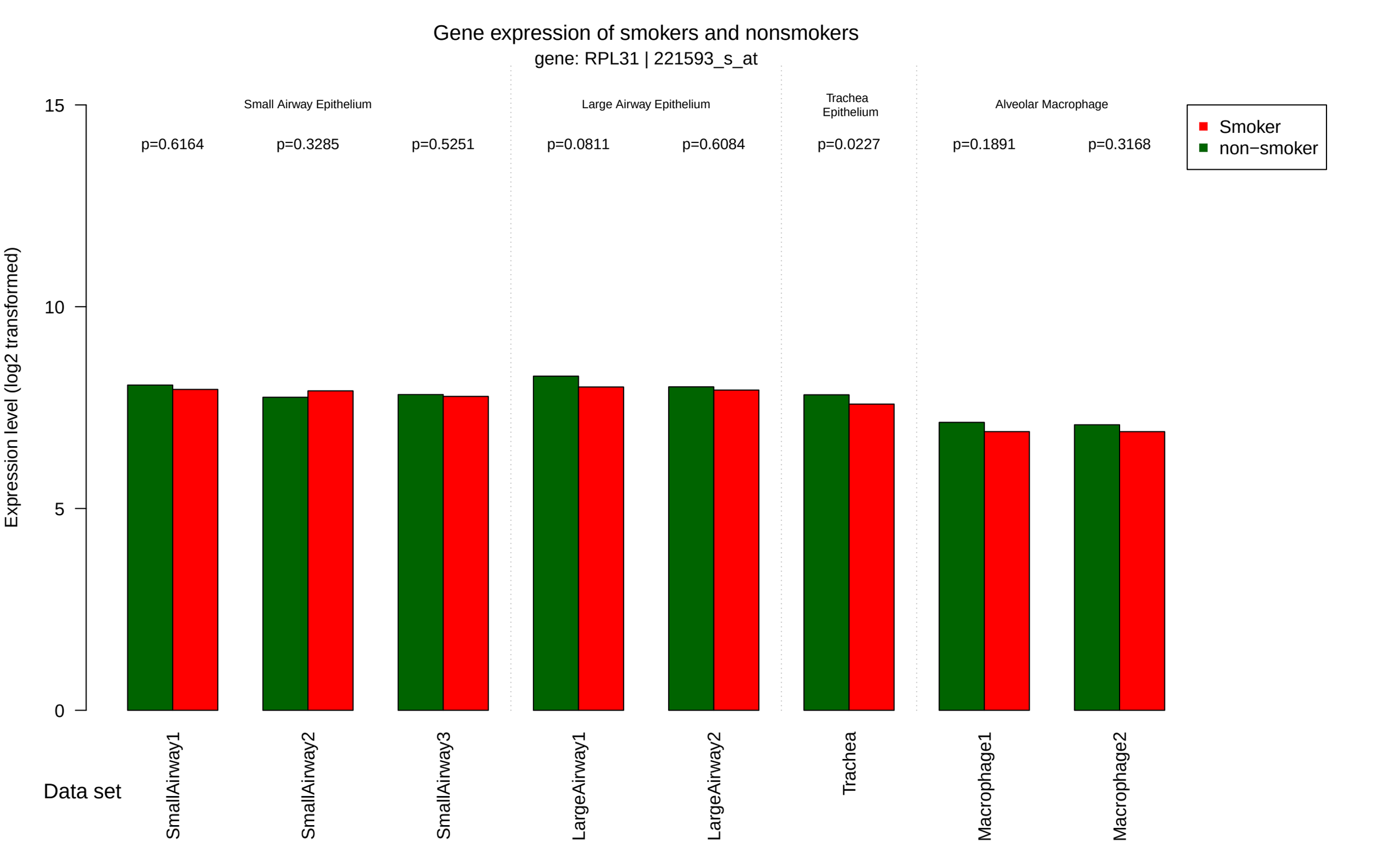
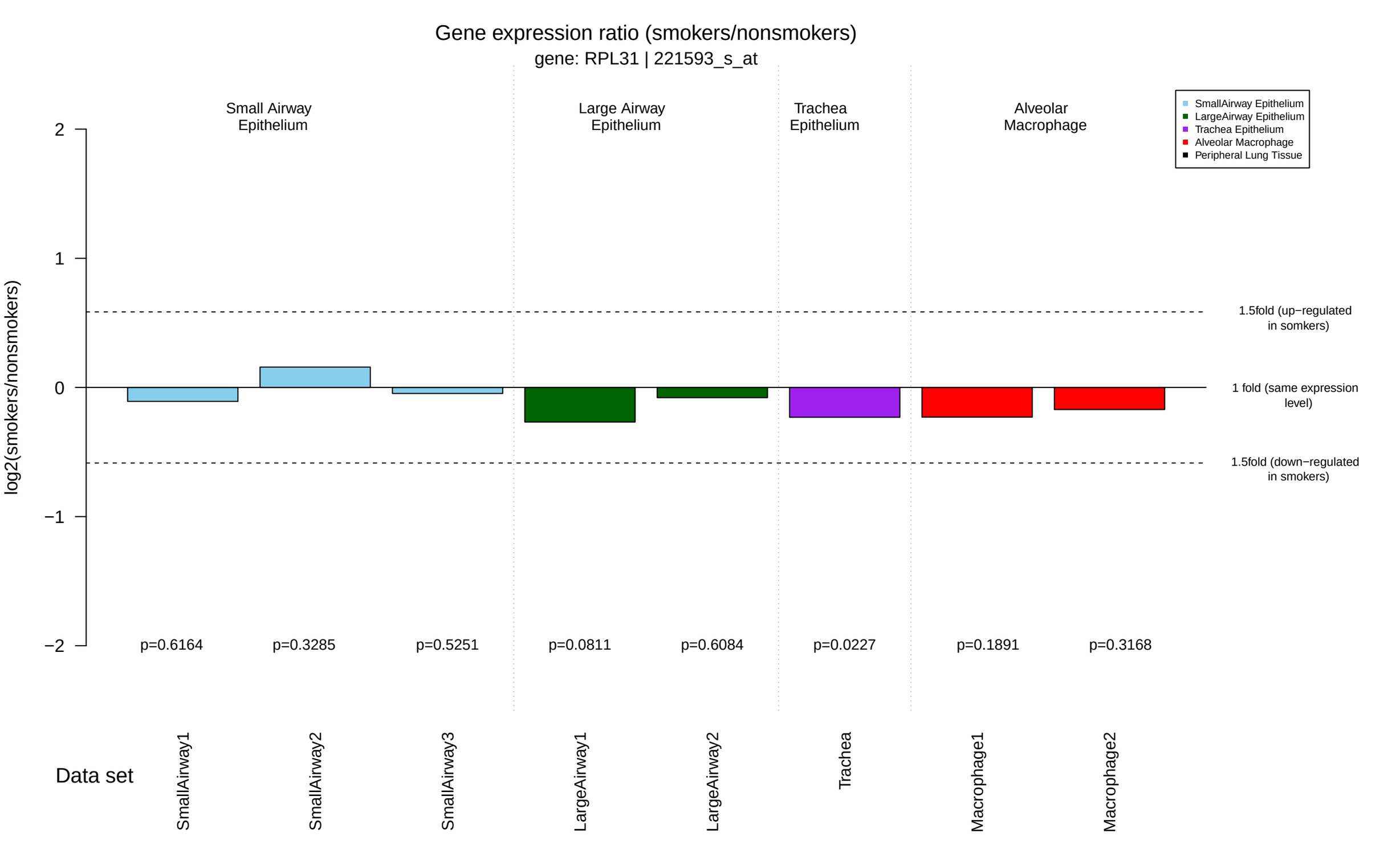
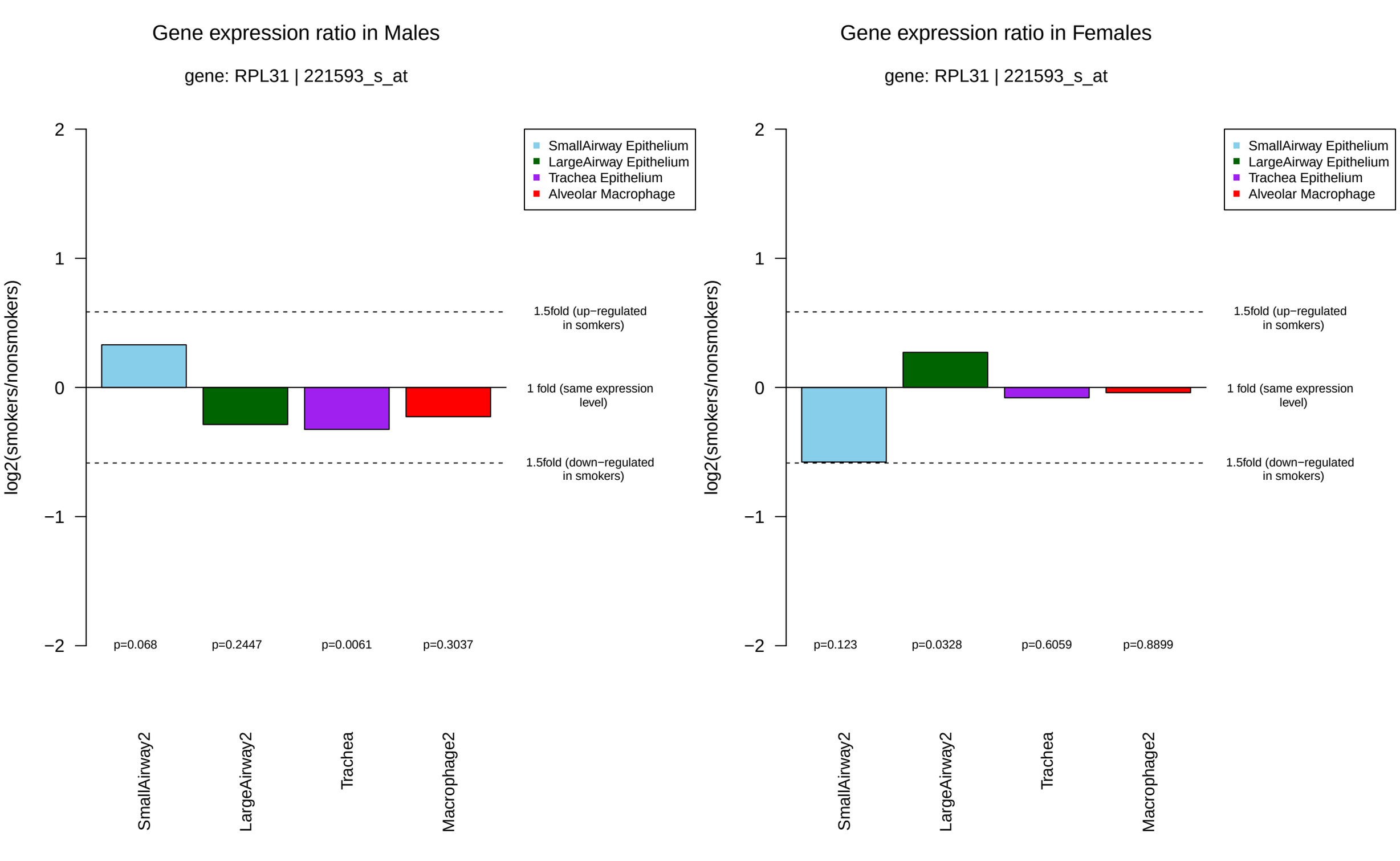
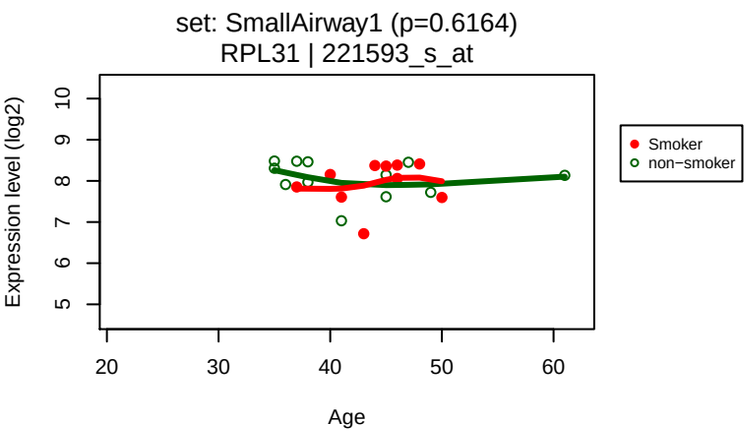
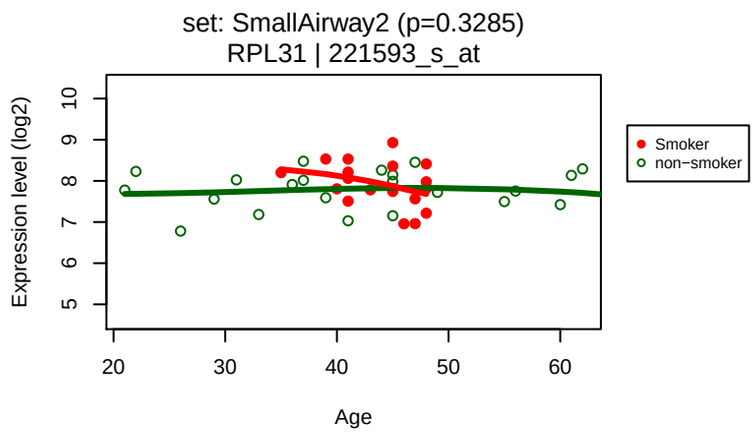
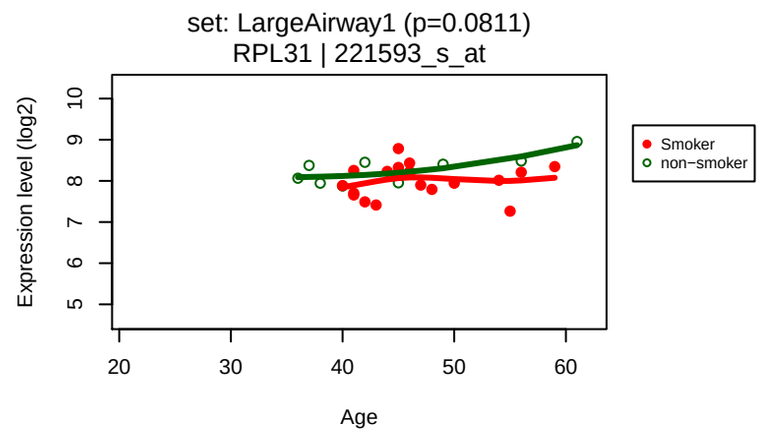

Gene expression of smokers and nonsmokers
gene: RPL31 | 221593_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6164
p=0.3285
p=0.5251
p=0.0811
p=0.6084
p=0.0227
p=0.1891
p=0.3168
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

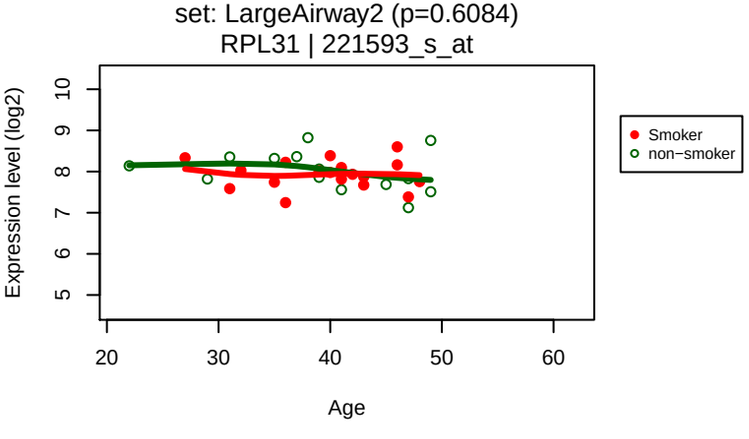

Gene expression ratio (smokers/nonsmokers)
gene: RPL31 | 221593_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6164
p=0.3285
p=0.5251
p=0.0811
p=0.6084
p=0.0227
p=0.1891
p=0.3168
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

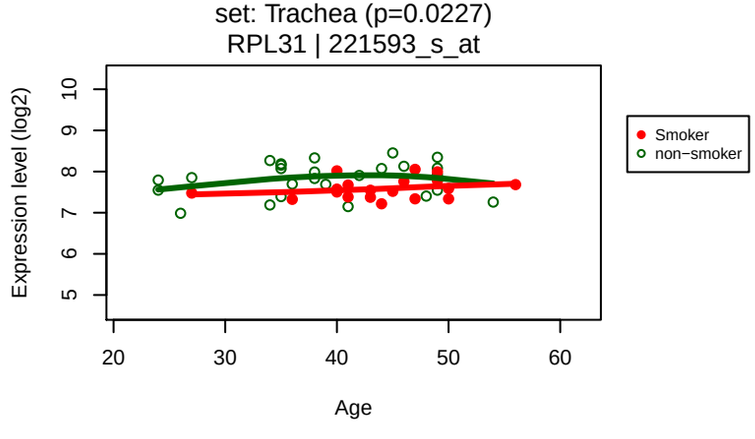

Gene expression ratio in Males
gene: RPL31 | 221593_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.068
p=0.2447
p=0.0061
p=0.3037
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPL31 | 221593_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.123
p=0.0328
p=0.6059
p=0.8899
SmallAirway2
LargeAirway2
Trachea
Macrophage2

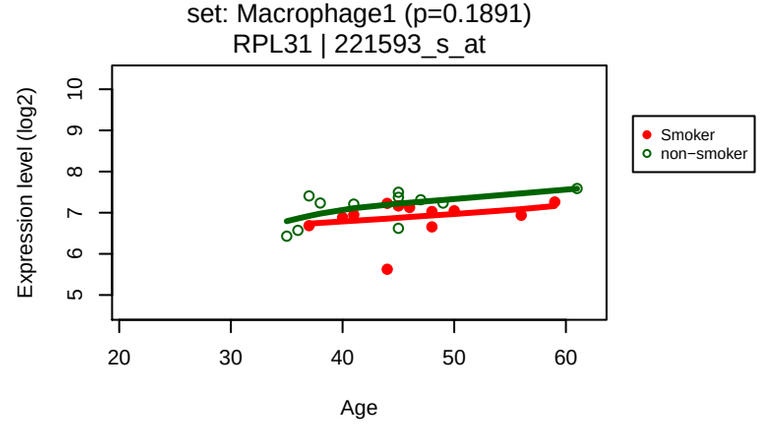

set: SmallAirway1 (p=0.6164)
RPL31 | 221593_s_at
Expression level (log2)
Age
Smoker
non−smoker

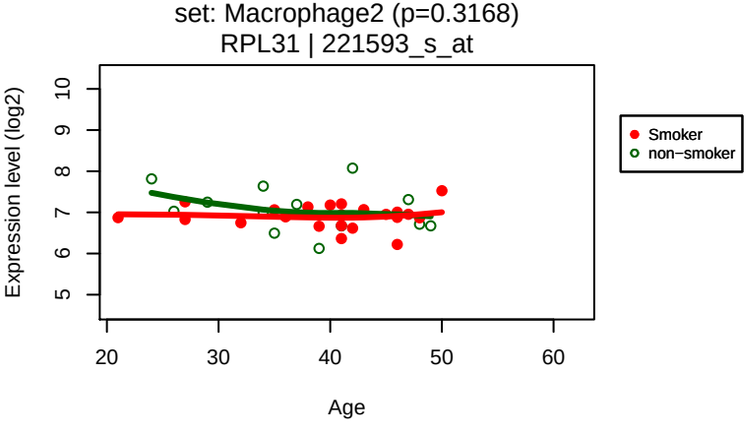

set: SmallAirway2 (p=0.3285)
RPL31 | 221593_s_at
Expression level (log2)
Age
Smoker
non−smoker

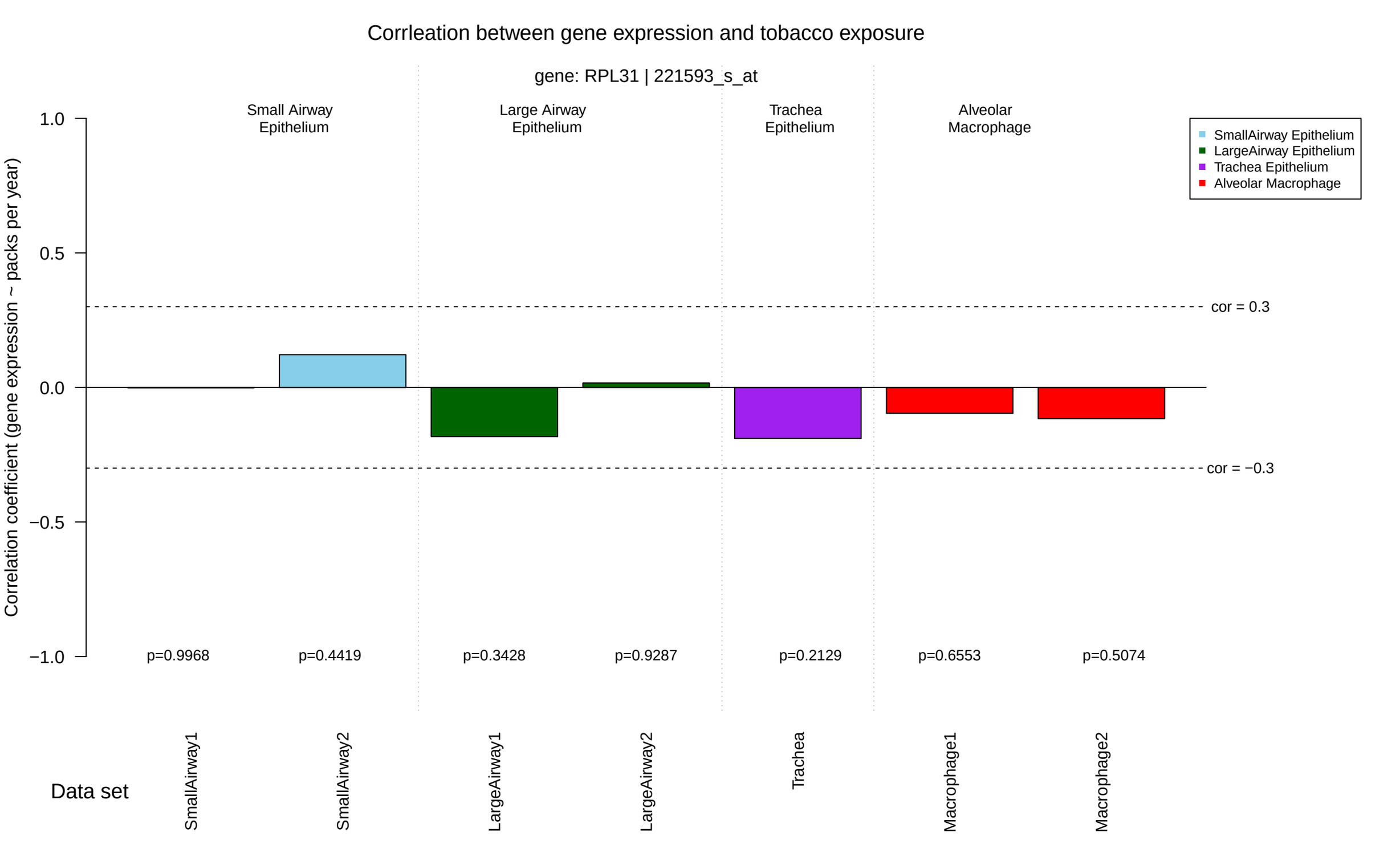

set: LargeAirway1 (p=0.0811)
RPL31 | 221593_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6084)
RPL31 | 221593_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0227)
RPL31 | 221593_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1891)
RPL31 | 221593_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3168)
RPL31 | 221593_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPL31 | 221593_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9968
p=0.4419
p=0.3428
p=0.9287
p=0.2129
p=0.6553
p=0.5074
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2